

Interactive Exploration of Subsets in Projections through Radial Attribute Fingerprints

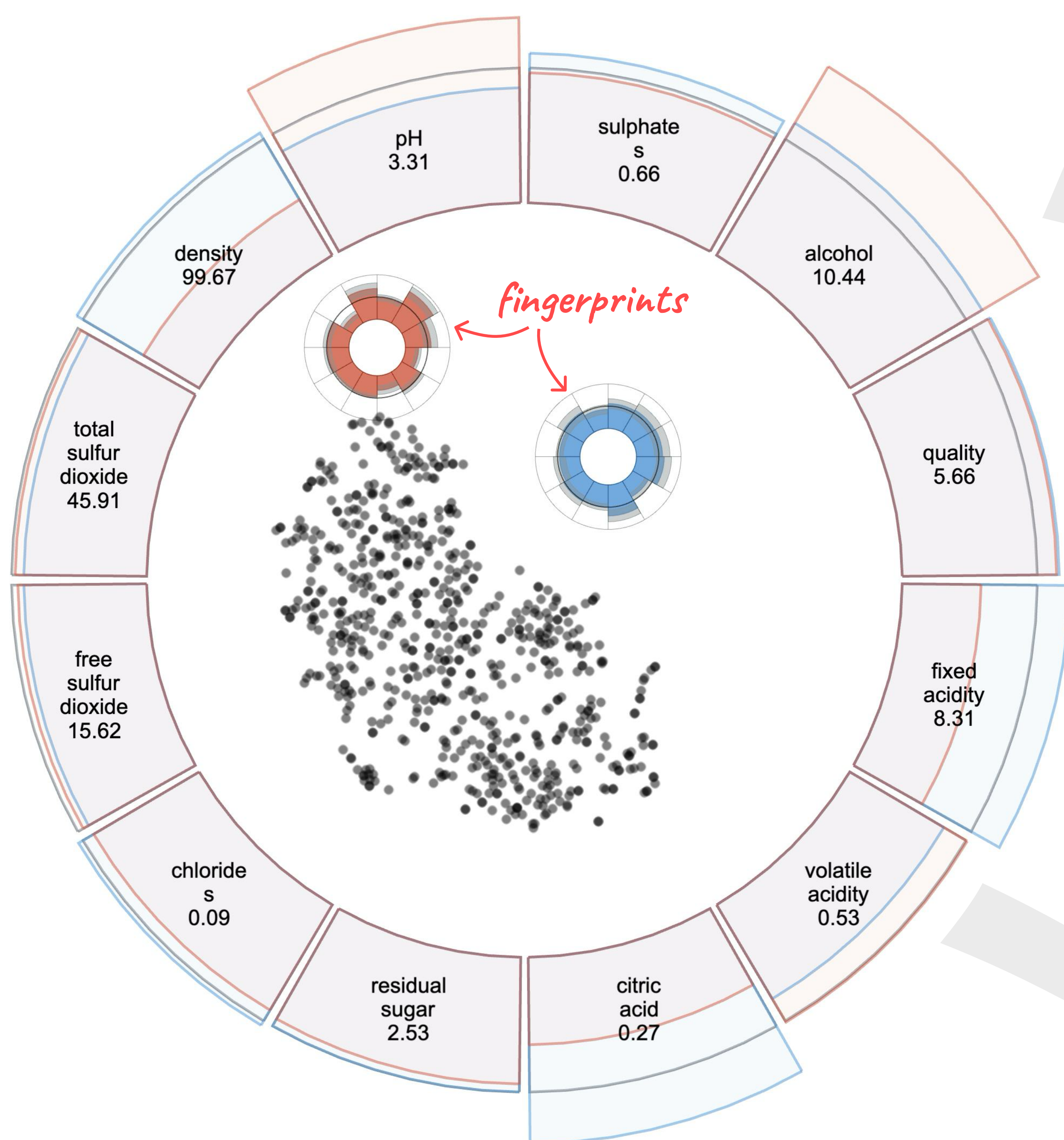
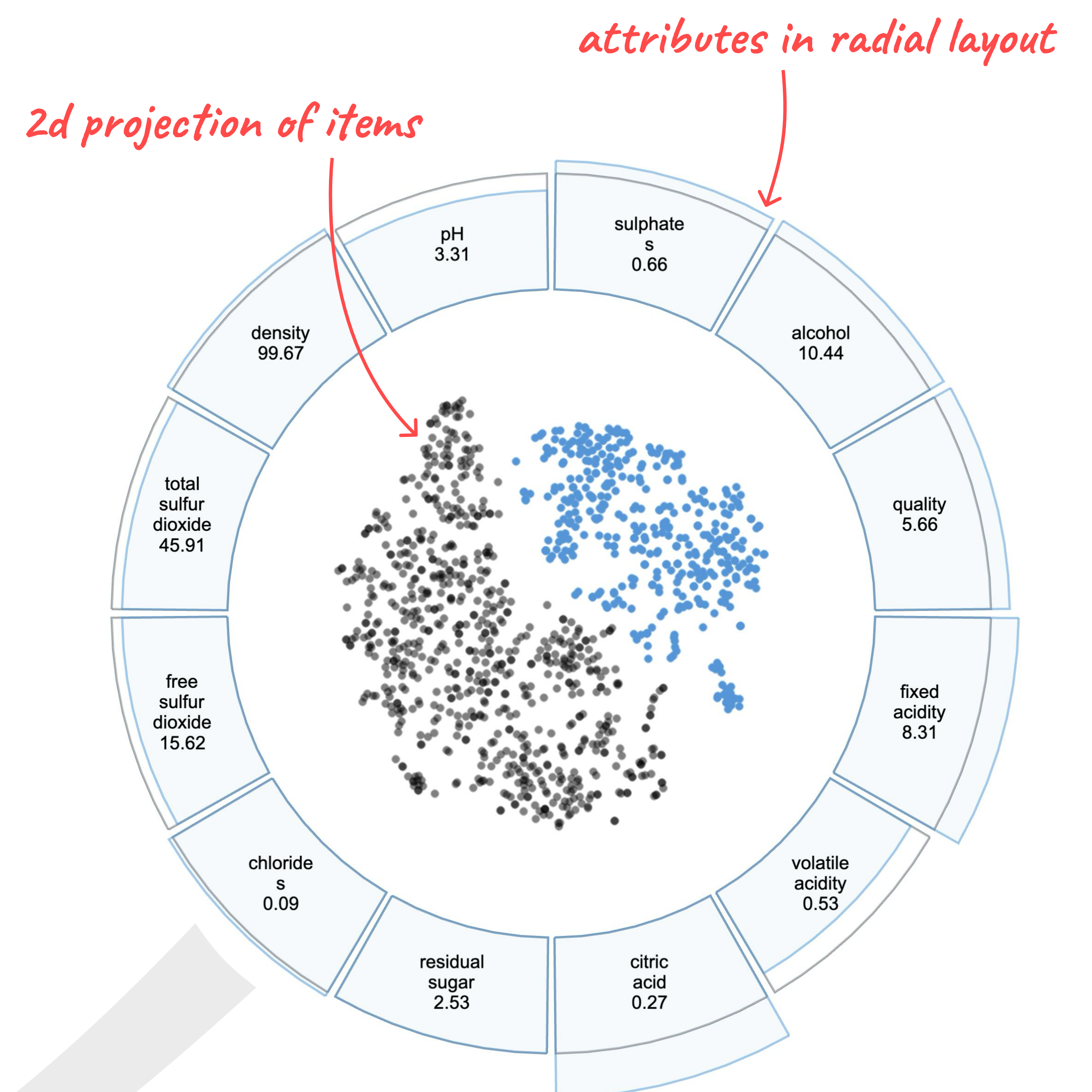
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PROBLEM

Two-dimensional projections reveal patterns such as clusters and outliers, but relating them back to the underlying high-dimensional attribute space remains challenging.

APPROACH

We introduce the visual metaphor of an attribute fingerprint, arranging high-dimensional attributes in a circular layout around the projection. The design enables users to create subsets, refine the exploration space through filtering and drill-down, and compare subsets based on their attributes.



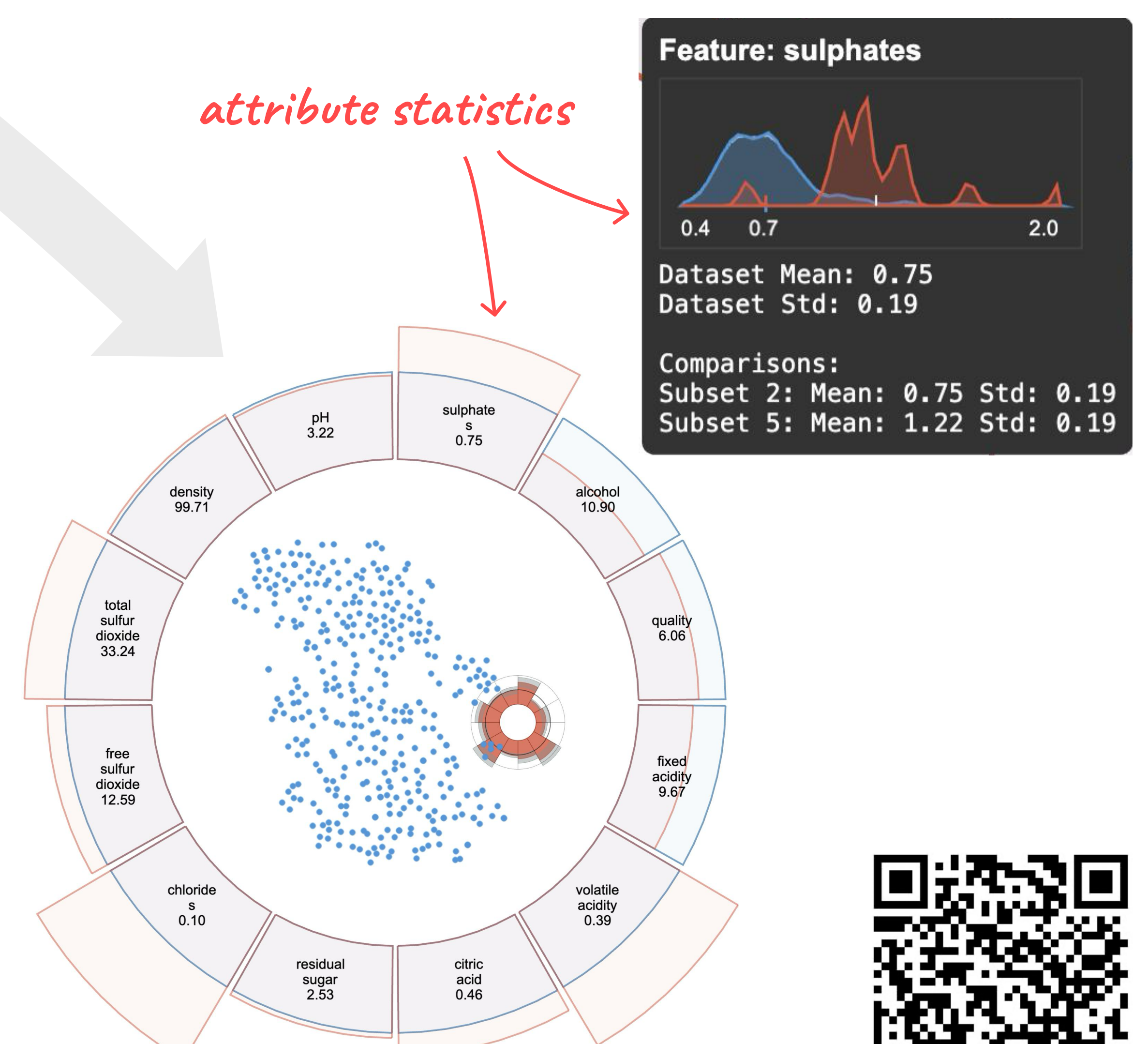
TASKS

- T1 Subset Creation** through ...
 - selection in projection space
 - algorithmic partitioning
 - attribute-based criteria
- T2 Subset Refinement** by ...
 - filtering items
 - filtering attributes
 - recursive drill-down into subsets
- T3 Subset Comparison** ...
 - of subset with global item space
 - across multiple subsets

REFLECTION

- + unified view for both items and attributes
- + visual and algorithmic subset creation possible
- + recursive drill-down into subsets
- ? users assumed attribute layout influences projection
- ? visual scalability issues in attribute ring

attribute statistics



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